

ALEXANDRIA INTL, LA, USA

WMO: 747540

Lat: 31.335N

Lon: 92.559W

Elev: 26

StdP: 101.02

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 93915

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.8	-1.2	-9.9	1.6	1.0	-7.4	2.0	3.2	10.6	12.0	9.1	14.0	2.2	10	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.1	36.2	24.9	35.0	24.9	33.9	24.8	26.9	31.9	26.4	31.8	26.0	31.3	2.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.1	28.7	25.1	20.2	28.4	24.6	19.6	28.1	83.9	31.8	82.0	31.8	80.4	31.4	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.5	7.6	6.7	DB	-6.1	38.1	2.1	1.8	-7.6	39.4	-8.8	40.4	-10.0	41.5	-11.5	42.8
			WB	-7.1	28.5	2.0	1.7	-8.5	29.8	-9.6	30.8	-10.8	31.8	-12.2	33.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	9.9	11.9	15.6	19.4	24.0	27.0	28.1	28.2	25.6	20.0	14.4	10.7
	DBStd	7.83	5.71	5.65	4.91	3.91	3.07	1.97	1.60	1.86	2.74	4.42	4.87	5.47
	HDD10.0	200	74	40	12	1	0	0	0	0	0	1	15	58
	HDD18.3	1021	267	190	112	33	3	0	0	0	1	35	136	244
	CDD10.0	3706	70	94	184	284	433	509	562	566	468	309	146	81
	CDD18.3	1485	4	11	27	67	177	259	303	307	219	86	17	8
	CDH23.3	14542	11	46	155	526	1627	2621	3231	3321	2077	779	120	29
	CDH26.7	6442	0	3	17	111	631	1227	1560	1680	946	251	16	1
Wind	WSAvg	2.7	3.3	3.4	3.4	3.4	2.9	2.3	1.9	1.9	2.2	2.4	2.7	3.1
Precipitation	PrecAvg	1518	142	124	140	116	128	124	113	105	103	115	144	164
	PrecMax	2220	354	335	335	375	303	350	338	265	302	408	445	528
	PrecMin	918	26	12	13	7	0	26	11	3	11	0	5	43
	PrecStd	294	84	66	77	87	73	78	66	61	62	101	84	86
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.8	27.1	28.3	31.2	34.0	37.0	37.2	38.0	36.2	32.7	28.7	26.2
		MCWB	19.1	19.7	19.8	22.1	23.4	24.7	25.4	24.9	23.7	23.4	21.4	21.1
	2%	DB	22.8	24.8	27.0	29.1	32.6	35.0	35.6	36.4	34.3	31.2	26.5	23.9
		MCWB	18.2	18.8	19.5	21.0	23.0	24.4	25.2	25.0	23.8	22.6	20.2	19.9
	5%	DB	21.3	22.8	25.1	27.8	31.6	33.6	34.3	35.1	33.0	29.4	24.4	22.2
		MCWB	17.6	18.2	18.7	20.4	22.9	24.5	25.4	25.1	23.8	21.4	18.9	19.1
	10%	DB	19.1	21.2	23.4	26.6	30.4	32.5	33.3	33.8	32.0	27.8	22.7	20.1
		MCWB	16.1	17.7	18.1	19.8	22.6	24.5	25.4	25.1	23.7	20.9	18.5	17.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.2	21.3	22.2	24.1	26.1	26.9	27.5	27.7	26.8	25.5	23.0	22.2
		MCDB	23.3	24.2	25.1	27.7	30.6	31.9	32.3	32.4	30.9	29.4	25.9	24.2
	2%	WB	19.5	20.4	21.1	23.0	25.0	26.2	26.8	26.8	25.8	24.5	21.8	20.8
		MCDB	21.9	23.4	24.4	26.5	30.0	31.5	32.1	32.1	30.7	28.5	24.3	23.3
	5%	WB	18.5	19.4	20.3	22.2	24.2	25.7	26.3	26.3	25.3	23.7	20.6	19.7
		MCDB	20.5	22.2	23.6	26.0	28.9	30.9	31.9	31.9	30.2	27.4	23.5	21.8
	10%	WB	16.9	18.1	19.3	21.4	23.6	25.3	25.9	25.9	24.7	22.7	19.2	17.9
		MCDB	18.8	20.5	22.5	25.0	28.1	30.5	31.2	31.3	29.3	25.9	22.0	19.6
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.1	11.6	11.8	11.3	10.5	10.3	11.1	11.7	13.1	12.5	10.9
		MCDBR	12.2	11.9	12.1	12.3	11.8	11.9	11.6	12.9	12.8	13.3	13.6	12.1
		MCWBR	8.3	7.0	6.3	5.6	4.1	3.7	3.6	3.7	4.3	5.7	8.0	8.4
	5% WB	MCDBR	9.7	10.1	10.4	10.0	10.0	10.1	10.3	10.9	10.4	10.3	11.3	10.7
		MCWBR	7.8	7.0	6.6	5.7	4.4	4.0	3.9	3.9	4.0	5.3	7.8	8.2
Clear-Sky Solar Irradiance	taub	0.331	0.352	0.372	0.413	0.440	0.479	0.500	0.494	0.443	0.372	0.355	0.343	
	taud	2.522	2.467	2.407	2.295	2.265	2.196	2.166	2.182	2.296	2.467	2.469	2.488	
	Ebn at Noon	897	908	910	881	855	817	798	799	828	873	859	861	
	Edh at Noon	87	100	114	132	137	147	150	146	126	99	91	86	
All-Sky Solar Radiation	RadAvg	2.76	3.15	4.38	5.41	6.01	6.06	6.02	5.68	5.01	4.26	3.18	2.44	
	RadStd	0.25	0.55	0.36	0.42	0.40	0.50	0.46	0.52	0.46	0.59	0.32	0.28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+118	+77

Nomenclature: See separate page